4 **(a)** Fig. 4.1 shows two identical small positive charged spheres A and B, each having a mass of 5.0 x 10^{-4} kg in equilibrium. Sphere A is fixed in position by an insulated rod while sphere B is free to move and attached to the ceiling by a string with length 0.20 m.

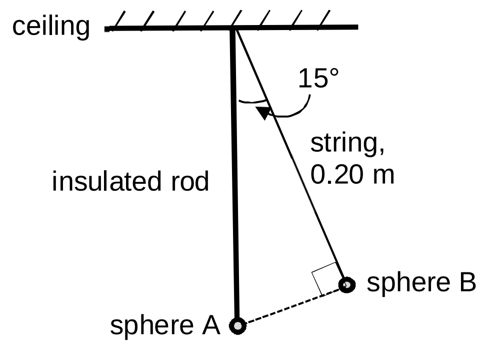


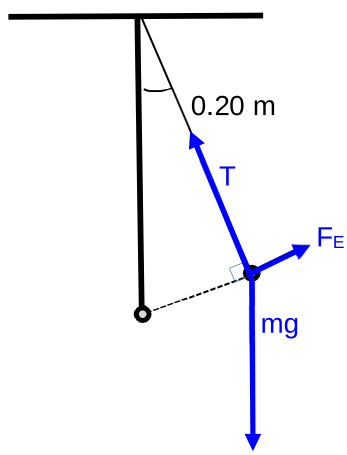
Fig. 4.1

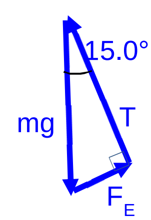
The string suspends at an angle of 15° to the vertical insulating rod and it also makes a right angle with the straight line connecting the two spheres.

Calculate

(i) the repulsive force acting on sphere B due to sphere A;

Draw FBD of sphere B:





Rearranging the vectors, we form a vector triangle

$$\sin 15° = \frac{F_E}{mg}$$

F_E = 5 x 10^{-4} x 9.81 x sin 15°
= 1.27 x 10^{-3} N
= 1.3 x 10^{-3} N (2 s.f.)

repulsive force = ………………… N [3]

(ii) the magnitude of the charge on each sphere.

By Coulomb's law: $F_E = q_1q_2/4\pi\varepsilon_o r^2 = q^2/\{4\pi\varepsilon_o[(0.20 \tan (15°)]^2\}$

$q = (0.20 \tan (15°))\sqrt{(4\pi(8.85 \times 10^{-12})(1.27 \times 10^{-3})}$
$= 2.01 \times 10^{-8}$ C
$= 2.0 \times 10^{-8}$ C (2 s.f.)

charge = …………………C [2]

(b) Sphere B connected with the same string as part **(a)** is now projected into the page at a speed of 0.50 m s^{-1} such that it orbits horizontally around the same vertical insulating rod as shown in Fig. 4.2.

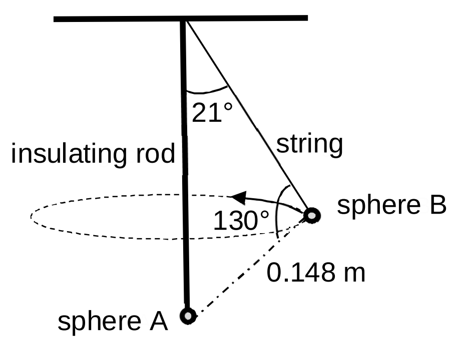


Fig. 4.2

The string makes an angle of 21° with the insulating rod. The distance between sphere A and sphere B is now found to be 0.148 m and the angle between the string and this line connecting sphere A and B is now 130°.

(i) Show that the resultant horizontal force on sphere B is 1.7 x 10^{-3} N.

The resultant horizontal force is the centripetal force,

$F_c = mv^2/r$
$= 5.0 \times 10^{-4} \times (0.50)^2/ (0.20 \sin 21°)$
$= 1.74 \times 10^{-3}$ N
$= 1.7 \times 10^{-3}$ N

[2]

(ii) Calculate the tension in the string.

The resultant horizontal force by the horizontal component of tension and the horizontal component of the electrostatic force provides the centripetal force.

The new electrostatic force

$= \frac{1}{4\pi\varepsilon_0}\frac{qq}{r^2}$

$= \frac{1}{4\pi(8.85\times10^{-12})}\frac{(2.01\times10^{-8})^2}{0.148^2}$

$= 1.642 \times 10^{-4}$ N

Hence,

$T \sin 21° - F_E \cos 61° = F_c$

$T \sin 21° - 1.642 \times 10^{-4} \cos 61° = 1.74 \times 10^{-3}$

$T = 5.1 \times 10^{-3}$ N (2 s.f.)

tension = ... N [3]

(c) A uniform magnetic field perpendicular to the motion of sphere B is now introduced to the system in part **(b)** as shown in Fig. 4.3.

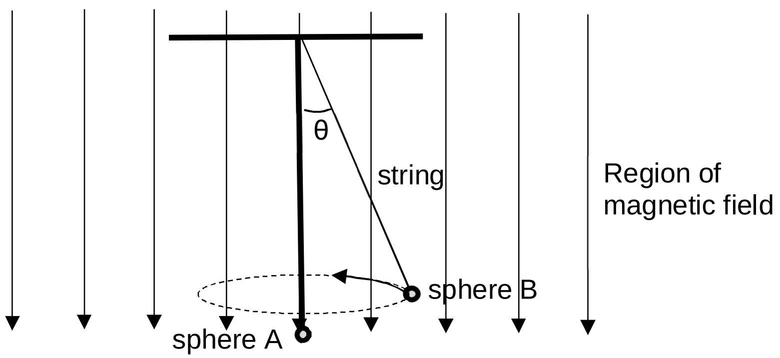


Fig. 4.3

Discuss how this might affect the subsequent motion of sphere B.

With the introduction of the magnetic field, there will be magnetic force directed horizontally towards the vertical insulating rod on sphere B.

The magnitude of the centripetal force will therefore increase.

Hence the speed of motion increase, and/or, the radius of circular motion will decrease.

[3]

(d) In the presence of air, sphere B and A will slowly discharge as they ionize the air particles around them.

Describe the path of sphere B when this happens.

It will spiral inwards / towards sphere A (and eventually come to a stop).

[1]

5 **(a)** State two differences between a progressive wave and a stationary wave.